



DATA SCIENCE PROJECT

Diabetes Risk Prediction

Building a Clinical Pre-Screening Tool with Machine Learning — from raw patient data to a live, interactive web app

768

Patient Records

3

Models Compared

88.31%

Test Accuracy

0.947

AUC-ROC

The Problem

Why early diabetes screening matters, and where ML fits in

Diabetes affects over 537 million adults worldwide.

Early detection dramatically improves outcomes, but full clinical testing is resource-intensive and not always accessible.

This project explores whether a model can flag high-risk patients using 8 easily-measurable clinical features — enabling faster, cheaper pre-screening.

Problem Framing

TYPE

Binary classification (Diabetic vs. Not)

DATASET

Pima Indians Diabetes Dataset (NIDDK)

SIZE

768 patients × 8 clinical features

POPULATION

Female patients, Pima Indian heritage, age 21+

TARGET USE

Pre-screening tool, not a diagnosis

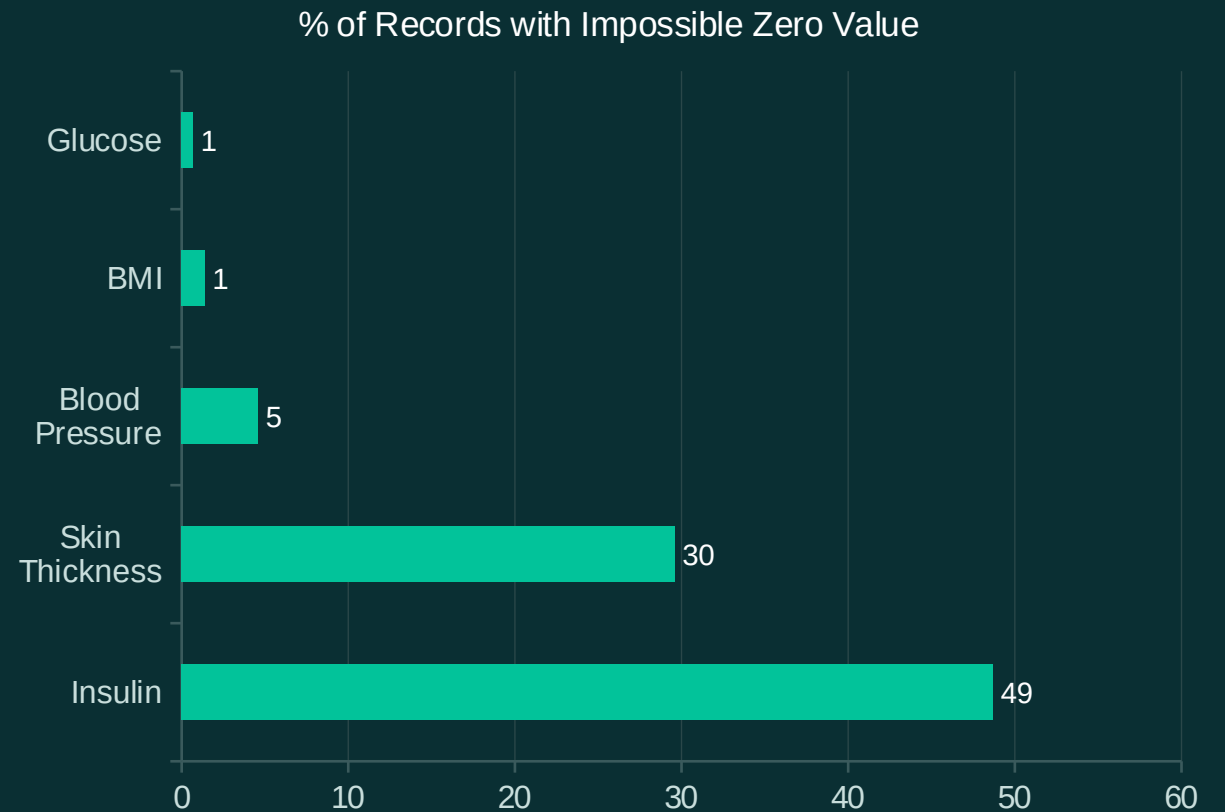


The Hidden Data Quality Trap

A 5-clinical-column problem that `.isnull().sum()` completely misses

Glucose, blood pressure, skin thickness, insulin, and BMI contained 0 values — biologically impossible for a living patient.

These weren't flagged as missing by standard checks — they were zeros hiding in plain sight. Catching this before modeling was the single most important cleaning step in the project.



Fix: zero → NaN, then median-imputed grouped by Outcome (not a blanket median)

End-to-End Pipeline

From raw CSV to a deployed prediction app

01

Clean

Zero-as-missing detection,
grouped-median imputation,
outlier flagging

02

Engineer

BMI categories, glucose risk
tiers, age groups added as
new features

03

Explore

Univariate + correlation
analysis to identify the
strongest predictors

04

Model

Logistic Regression, Random
Forest & XGBoost, 5-fold
cross-validated

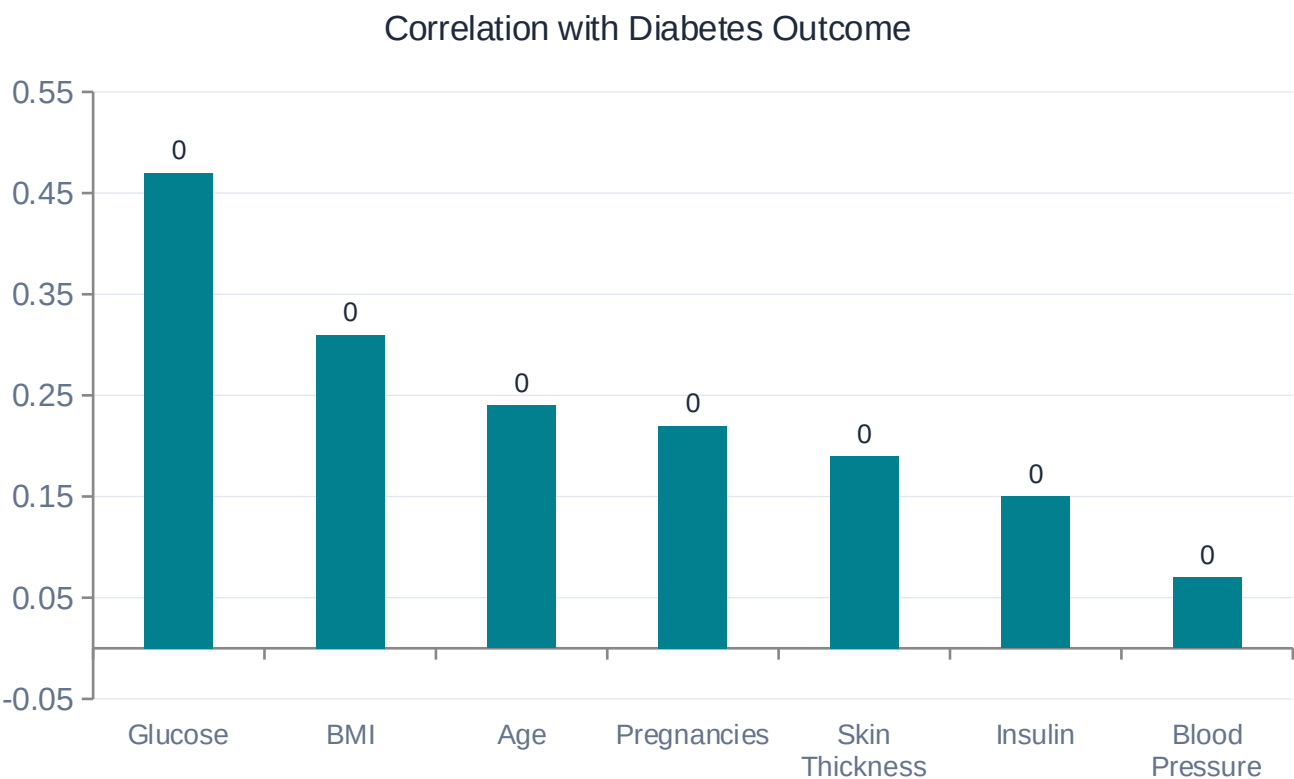
05

Deploy

Best model serialized with
pickle and served via a
Streamlit app

What the Data Reveals

Correlation analysis surfaces the real risk drivers



Glucose shows the clearest raw signal

Diabetic patients averaged ~141 mg/dL vs. ~110 mg/dL for non-diabetics — the clearest univariate signal (though the trained model weighs Insulin more heavily overall — see slide 8).

BMI & family history compound

BMI correlates ~0.65 with skin thickness — both proxy body composition risk.

Blood pressure underperforms

Weakest predictor in the set — a reminder that intuition and data don't always agree.

Modeling & Validation

Three algorithms, rigorously compared — not just one lucky split



Logistic Regression

Baseline, fully interpretable coefficients



Random Forest

Captures non-linear feature interactions



XGBoost

Gradient-boosted trees for stronger recall

Validation approach

5-fold cross-validation

on every model — a single 80/20 split can be lucky or unlucky on a dataset this size.

Stratified splitting

to preserve the 65/35 class balance in every fold.

Metric focus: Precision, Recall & AUC-ROC for the diabetic class — accuracy alone is misleading on imbalanced data.



The accuracy trap

65.1%

of patients don't have diabetes. A model that predicts "no diabetes" for everyone scores 65% accuracy — and is completely useless.

Results

Best model: XGBoost — highest accuracy, AUC-ROC, and recall of the three models tested

88.31%

Accuracy

0.947

AUC-ROC

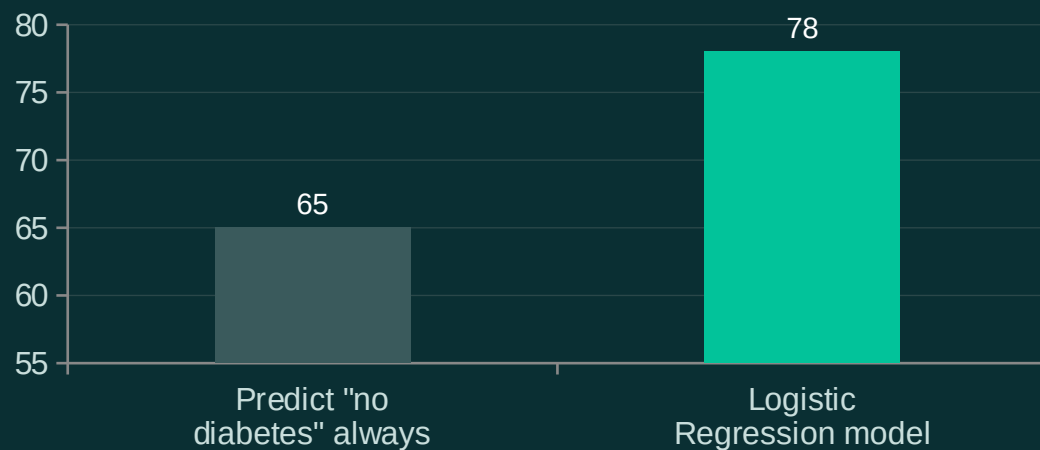
82.14%

Precision (diabetic)

85.19%

Recall (diabetic)

Baseline vs. model



Why recall matters most here

In a pre-screening tool, a missed diabetic (false negative) is far more costly than a false alarm. XGBoost's 85.19% recall is a substantial improvement over the Logistic Regression baseline's 55.56% — but continued improvement remains valuable given the cost of a missed case.

What Drives the Prediction

XGBoost feature importance, ranked by impact

1

Insulin

The single strongest predictor at ~46% importance — despite the highest missingness pre-imputation

2

SkinThickness

Body-fat proxy — second-strongest signal at ~11%

3

Glucose

The classic diabetes marker, but ranks third here at ~10.5% — not the dominant feature

4

Age

Risk increases with age, ~10% importance

5

BMI

Obesity is a well-established risk factor, ~6.5% importance



Clinical takeaway

Insulin drives the largest share of the prediction — a counterintuitive finding worth flagging for clinical review, since Glucose is the biomarker most associated with diabetes in established literature.

Blood pressure alone is a weak signal — it needs to be paired with other markers, not used in isolation.

This mirrors real clinical research on Type 2 diabetes risk factors.



Shipping It: Live Deployment

From a Jupyter notebook to a public, interactive web app



Model trained
& validated



Serialized
with pickle



Pushed to
GitHub



Deployed on
Streamlit Cloud



Live on
portfolio site

Try it yourself



Live App

[Insert Streamlit URL]



Source Code

github.com/yemifatodu/diabetes-prediction-app



Portfolio

yemifatodu.online

Tech Stack

Tools used across the full pipeline

Language & Data

Python • Pandas • NumPy

Visualization

Matplotlib • Seaborn • Plotly

Machine Learning

Scikit-learn • XGBoost • GridSearchCV

App & Deployment

Streamlit • Streamlit Cloud • Pickle

Environment

Jupyter Notebook • Anaconda • VS Code

Version Control

Git • GitHub

What I'd Improve Next

Treating this as a living project, not a one-off submission

High priority

- Add a precision-recall curve to complement the existing confusion matrices and ROC curve
- SMOTE or `class_weight='balanced'` for class imbalance
- Grouped-median imputation fully documented in README

Makes it stand out

- Hyperparameter tuning via GridSearchCV
- SHAP values for model explainability
- Expand the Streamlit form with confidence scores

Nice to have

- A/B the three models live in the app
- Add a bank churn project as a second case study
- Automated retraining pipeline via GitHub Actions

Let's Talk Data

This project shows the full lifecycle — messy data, honest evaluation, and a real deployed product. Open to feedback, collaboration, and opportunities.



Portfolio

yemifatodu.online



Live Demo

[Insert Streamlit URL]



GitHub

github.com/yemifatodu/diabetes-prediction-app